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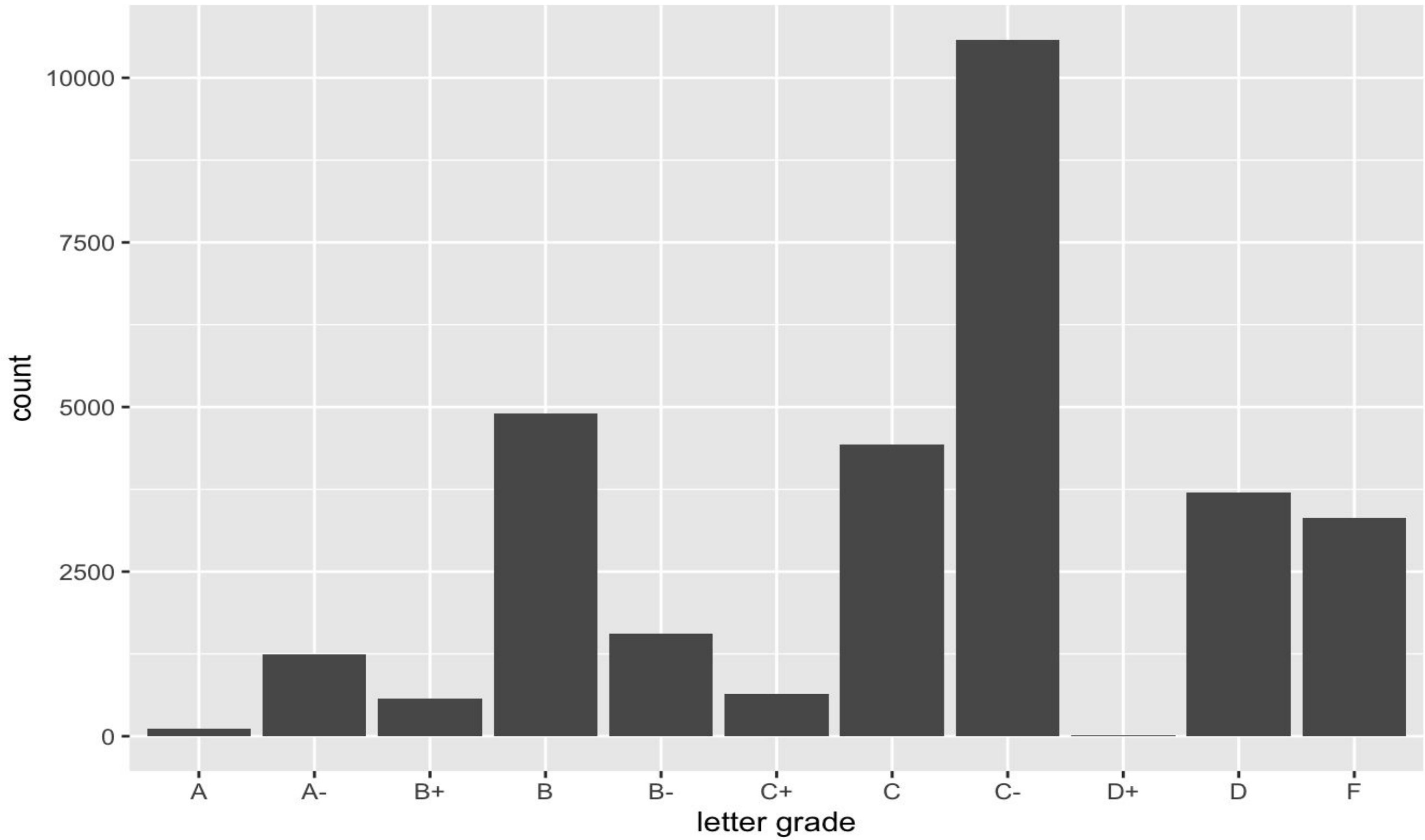
Ex-Ante Cybersecurity Risk

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Research Question: How adequate are firms in their ex-ante cybersecurity measures?

Cybersecurity Score A-F Scale



This bar graph, generated by an AI model, visualizes the distribution of cybersecurity scores across a large dataset, where the scores (A through F) are determined by the presence and effectiveness of various cybersecurity measures. The AI was used to extract and map relevant cybersecurity information from 10-K filings, supporting our proprietary cybersecurity risk assessment framework to help generate preliminary evaluations. However, these assessments are based on our established methodology and require validation against actual cybersecurity performance data.

Background

With cybersecurity and technological developments rapidly transforming modern society, their presence in daily life has become increasingly critical, with finance at the forefront due to its reliance on data protection, risk management, and decision-making precision. As both corporations and cyber threats advance their methods, the implementation of robust cybersecurity measures has become essential for maintaining efficiency, resilience, and trust in financial systems. Thus, evaluating the effectiveness of emerging security programs and digital currency safeguards is necessary to strengthen financial stability, protect transactions, and guide future innovation.

Project Goal(s)

This research aimed to analyze how firms proactively managed cyber risk before a breach occurred. We used advanced Large Language Models (LLMs) to parse and categorize discussions of cybersecurity preparedness found in publicly available corporate disclosures, such as SEC filings. The project examined how these disclosures reflected a company's financial adjustments, insurance coverage, and security infrastructure in anticipation of potential cyberattacks.

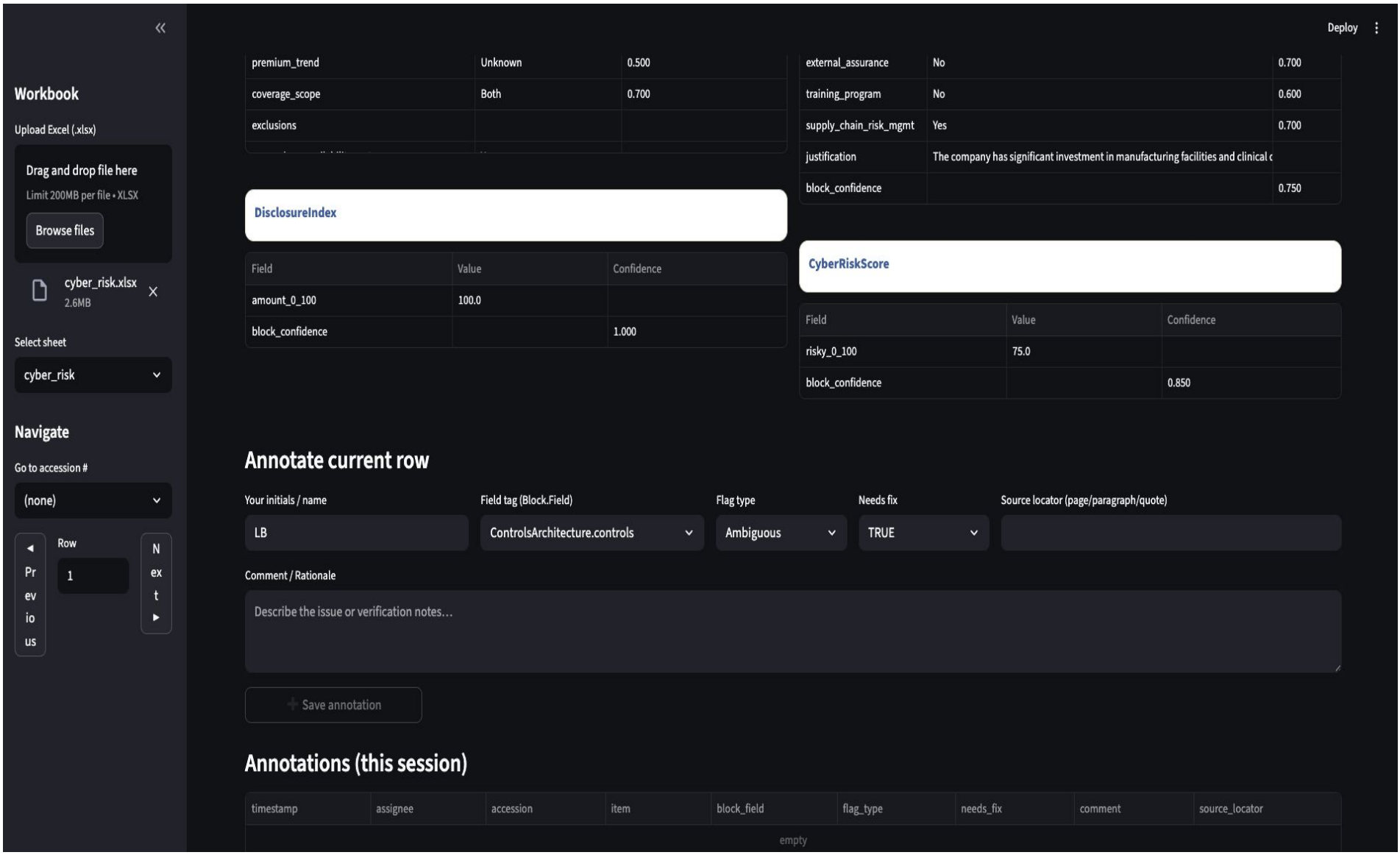
Data

Our Research uses a comprehensive dataset combining structured and unstructured data from a variety of sources. We leveraged official filings from the SEC, including 10-K, 8-K, 1A, and 1C filings, along with formal letters from the FDIC. To capture unstructured information, we used an AI model to scan 1A and 1C filings, which helped us efficiently extract key data on cybersecurity measures. We also analyzed raw communications between hackers and companies, providing real-world context on cyberattacks. Finally, we incorporated articles from the SEC to understand regulatory expectations for cybersecurity disclosures. With diverse data types, our project provides a holistic view, connecting company disclosures with real-world factors.

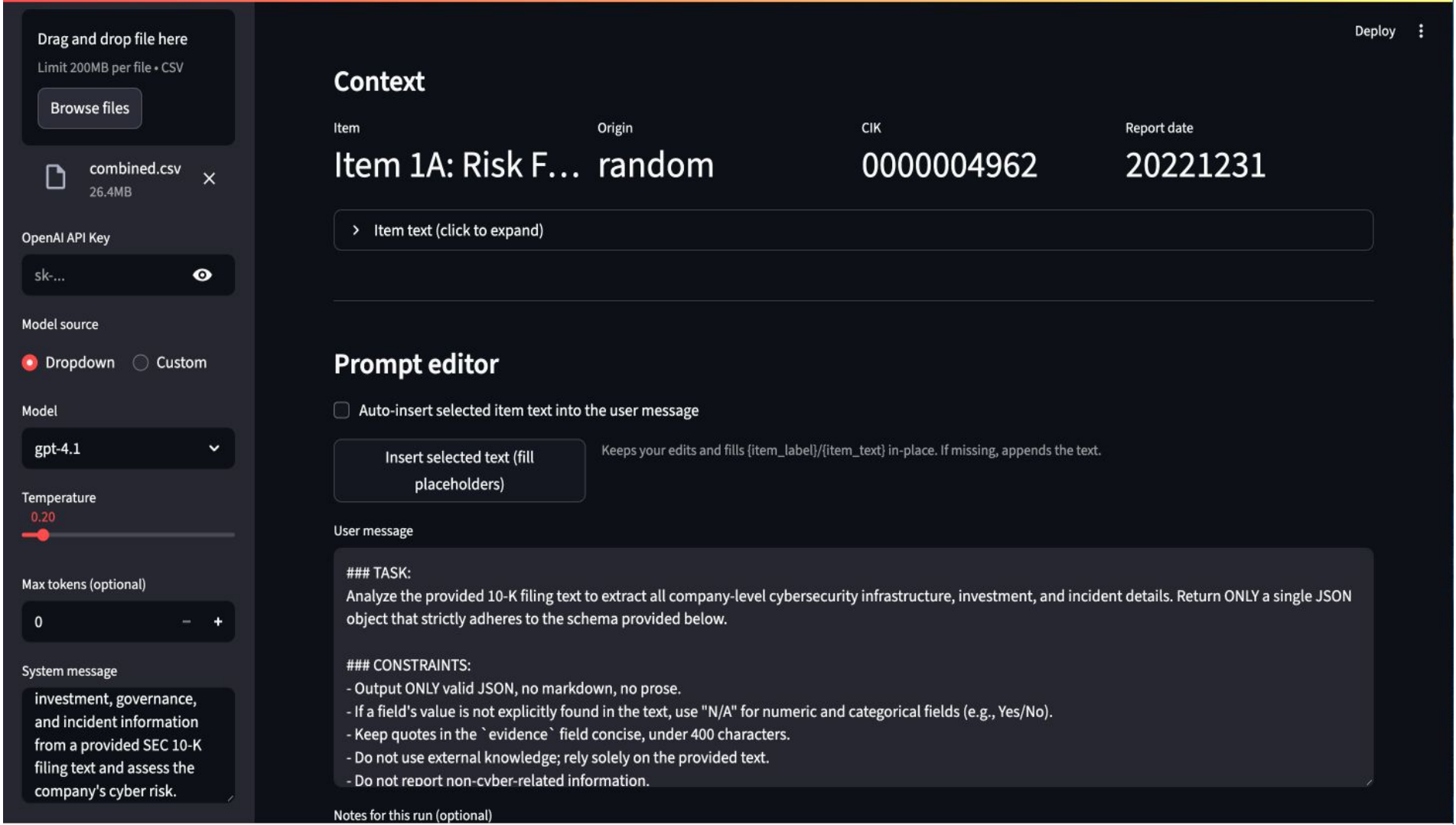
AI Integration

Artificial Intelligence has rapidly integrated into the finance industry, transforming operations across banking, analytics, and research. Financial institutions now employ AI assistants and analyst tools to streamline services, enhance security, and improve decision-making precision. In research, AI has become essential for keeping pace with modern developments. Large language models (LLMs) and AI-based applications accelerate due diligence and shift the focus toward prompt engineering and advanced analytical methods, reducing reliance on traditional approaches.

Research Devices



Our research used the custom-built Cyber Risk RA Annotator, a Python app created with Streamlit, to validate the accuracy of cyber risk data extracted by Large Language Models (LLMs) from corporate disclosures. The app allowed researchers to navigate through SEC filings, viewing the LLM-generated data in a structured, user-friendly format. This facilitated a field-by-field verification of the model's output against the source text. This human-in-the-loop approach was essential for confirming the reliability of the extracted data and identifying areas where the LLMs needed improvement.

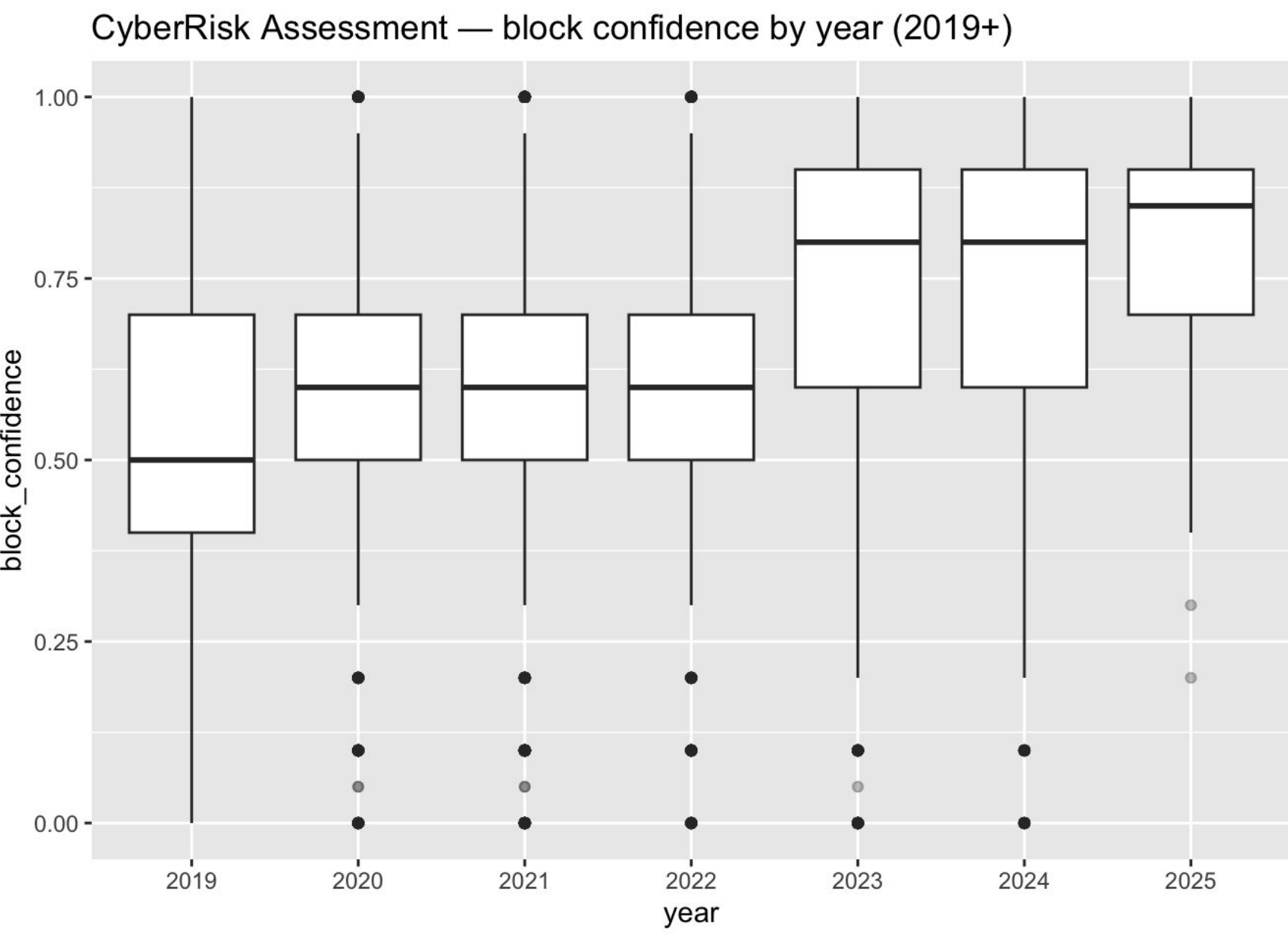


Our research used the custom-built Cyber Risk Prompt Lab, a Python app created with Streamlit, to refine how Large Language Models (LLMs) analyze cybersecurity disclosures. The app allowed researchers to systematically experiment with different prompts, models, and parameters to optimize the extraction of accurate data from SEC filings. By logging each attempt, we could identify the most effective prompts for specific tasks. This iterative, human-in-the-loop process was crucial for improving the reliability of our LLM-based analysis of how firms manage cyber risk, enabling us to better understand how companies proactively manage these risks.

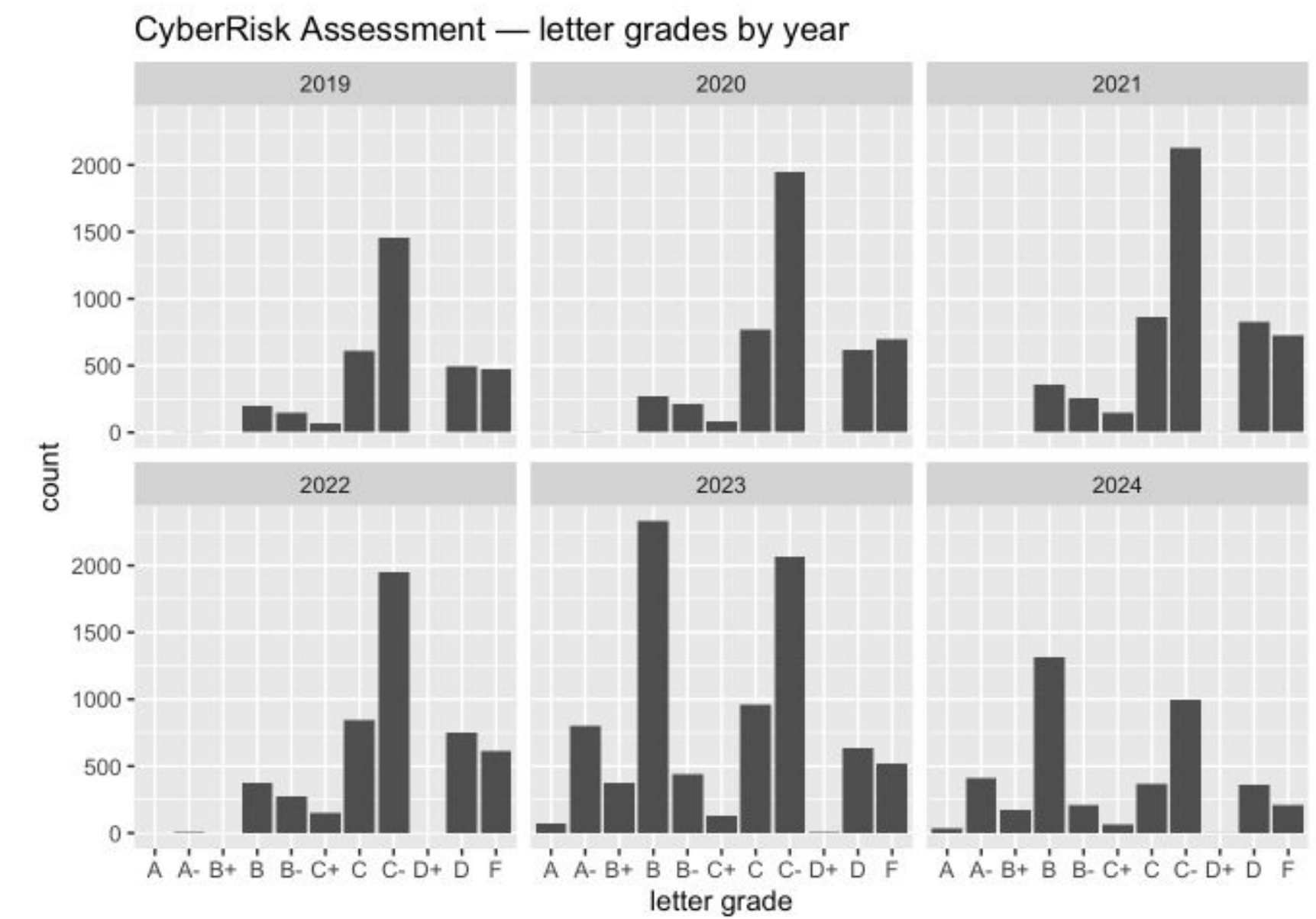
Findings:

While most firms have established baseline cybersecurity measures, our results reveal wide variability in overall preparedness. A large share of firms fall into mid-level ratings, reflecting partial compliance with security frameworks but limited proactive risk management. Only a smaller group demonstrate robust, well-integrated protections that signal strong resilience against evolving threats. Our preliminary analysis also suggest there may be a disconnect between what firms disclose and the actual strength of their cybersecurity infrastructure. This gap suggests that, despite progress, many organizations remain vulnerable and highlights the need for stronger, forward-looking safeguards to ensure resilience, trust, and long-term stability in financial systems.

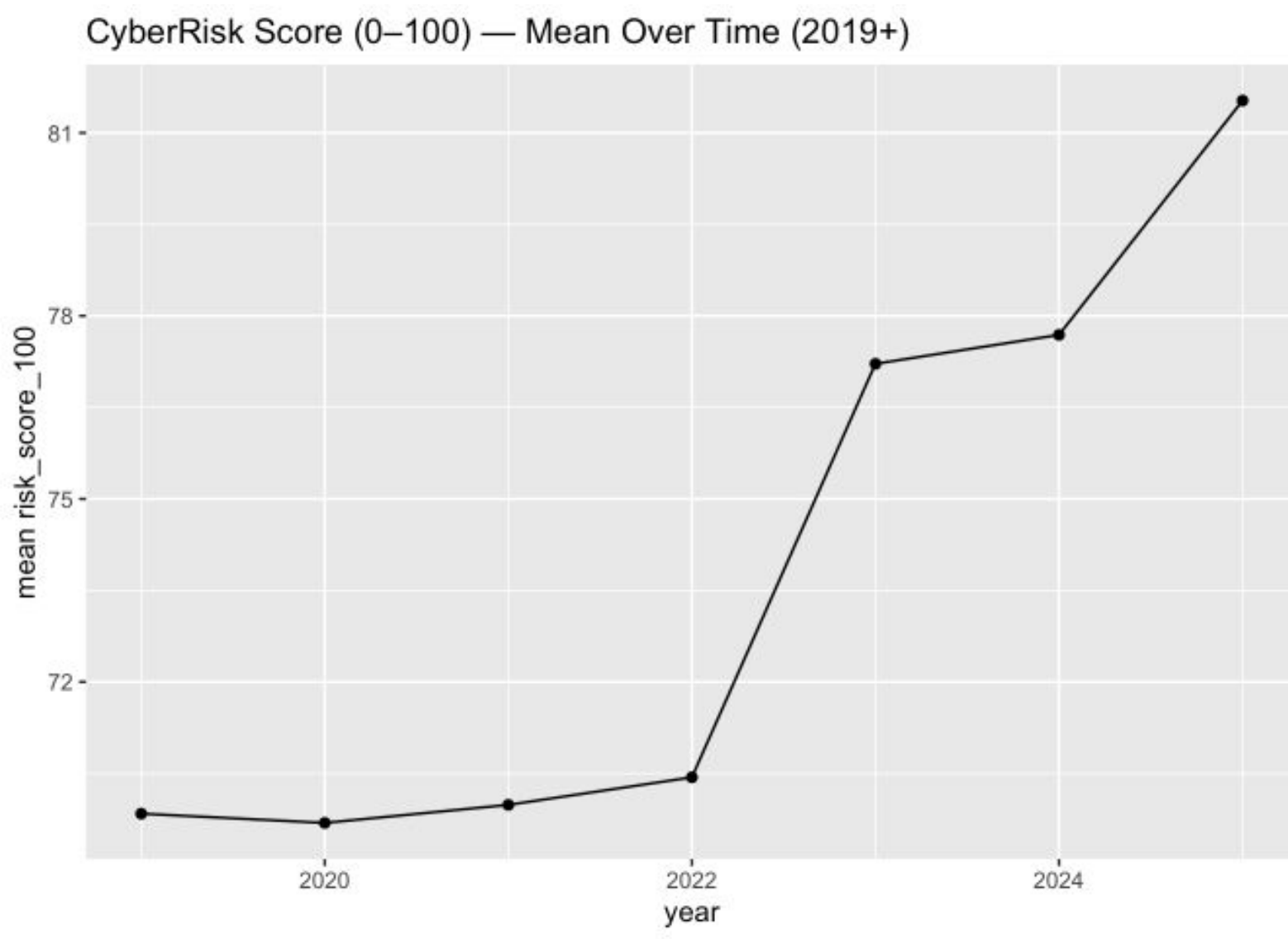
Block Confidence



Yearly Distribution



CyberRisk Score/Time



References

- <https://www.sec.gov/>